Check for updates

JGR Planets

RESEARCH ARTICLE

10.1029/2025JE009609

The Length of Martian Crater Rays and Their Relation to Lunar Cold Spots

Trevor P. Erwin[1], Brandon C. Johnson[1,2], David Minton[1,2], and Alexandria V. Johnson[1]

[1]Department of Earth, Atmospheric, and Planetary Sciences, Purdue University, West Lafayette, IN, USA, [2]Department of Physics and Astronomy, Purdue University, West Lafayette, IN, USA

Key Points:

- Martian crater rays are longer than lunar crater rays by an order of magnitude
- Lunar cold spots have ray lengths similar to Martian crater rays
- We propose a relationship between the formation mechanisms of lunar cold spots and Martian crater rays

Correspondence to:

T. P. Erwin,
erwin19@purdue.edu

Citation:

Erwin, T. P., Johnson, B. C., Minton, D., & Johnson, A. V. (2026). The length of martian crater rays and their relation to lunar cold spots. *Journal of Geophysical Research: Planets*, *131*, e2025JE009609. https://doi.org/10.1029/2025JE009609

Received 8 DEC 2025
Accepted 23 JUL 2026

Abstract Impact-generated crater rays are well-documented on the Moon, with most appearing as high-albedo streaks extending radially from a crater's center. On Mars, however, crater rays are significantly rarer and discernible only through thermal imaging due to their lower thermal inertia compared to surrounding terrain. This study presents the first comparative analysis between the lengths of Martian and lunar crater rays, including lunar albedo rays and cold spots, which are ray-like thermal anomalies associated with many of the youngest lunar craters. Our findings indicate that both Martian crater rays and lunar cold spots extend significantly farther than lunar albedo rays, with lengths an order of magnitude greater for craters of equivalent diameter. Furthermore, we propose a connection between the formation mechanisms of Martian crater rays and lunar cold spots based on their thermal properties. By integrating thermal rays into existing ejecta models, we refine the understanding of crater-ray formation and suggest that Martian crater rays and lunar cold spots may share a similar formation mechanism via secondary cratering processes. Advancing knowledge of these features has implications for impact dynamics and surface evolution across planetary bodies.

Plain Language Summary Craters on both the Moon and Mars can form rays, which are streaks extending radially from a crater's center. Lunar rays are much more common, and often have bright albedo features, appearing lighter than surrounding material. Martian rays, however, are much rarer, and are only visible in thermal imaging due to their low thermal inertia compared to background material. This work compares the lengths of Martian and lunar crater rays for the first time, as well as lunar cold spots, which, like Martian crater rays, are often only visible using thermal imaging due to their lower thermal inertia. This work finds that both Martian crater rays and lunar cold spots are longer than albedo lunar rays by an order of magnitude for craters of the same diameter. In addition, we propose that the mechanisms for the formation of Martian crater rays and lunar cold spots may be linked due to their unique thermal properties. Properly understanding the formation of these thermal rays will lead to a better understanding of impact dynamics, and could help to redefine how we measure ejecta patterns on different bodies.

1. Introduction

Crater rays are a common feature of rocky bodies in our solar system. These rays are streaks of material with bright albedo features that extend radially from fresh craters (Melosh, 1989). Their lengths vary, but longer ones, such as those surrounding Tycho crater on the Moon, span nearly an entire hemisphere (Baldwin, 1963). Crater rays are observed on the Moon (Baldwin, 1963), Mercury (Neish et al., 2013), Mars (McEwen et al., 2005), and the moons of Jupiter (Schenk & McKinnon, 1991). Before their discovery on Mars by McEwen et al. (2005), it was widely believed that atmospheric processes would prevent crater ray formation (Melosh, 1984). However, more than 100 rayed craters have since been identified on the Martian surface (Harris et al., 2023), though unlike their lunar counterparts, Martian crater rays are primarily visible in thermal data (McEwen et al., 2005).

Lunar rays have been classified into two categories; immature rays and mature rays (Hawke et al., 2004). Immature rays form immediately upon impact, consisting of freshly excavated high-albedo material that creates bright streaks across the lunar surface. Over time, space weathering darkens these rays, transitioning them into mature rays. Mature rays are often darker, but some can sill appear bright due to compositional differences, where the ejected material has a higher intrinsic albedo than the surrounding terrain. These mature rays will eventually darken to the point where they are no longer distinguishable from the background regolith. Viewing conditions also affect ray visibility, with low-incidence angles enhancing contrast on both the Moon and Mercury (Neish et al., 2013).

© 2026 The Author(s). Journal of Geophysical Research: Planets published by Wiley Periodicals LLC on behalf of American Geophysical Union.
This is an open access article under the terms of the Creative Commons Attribution License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

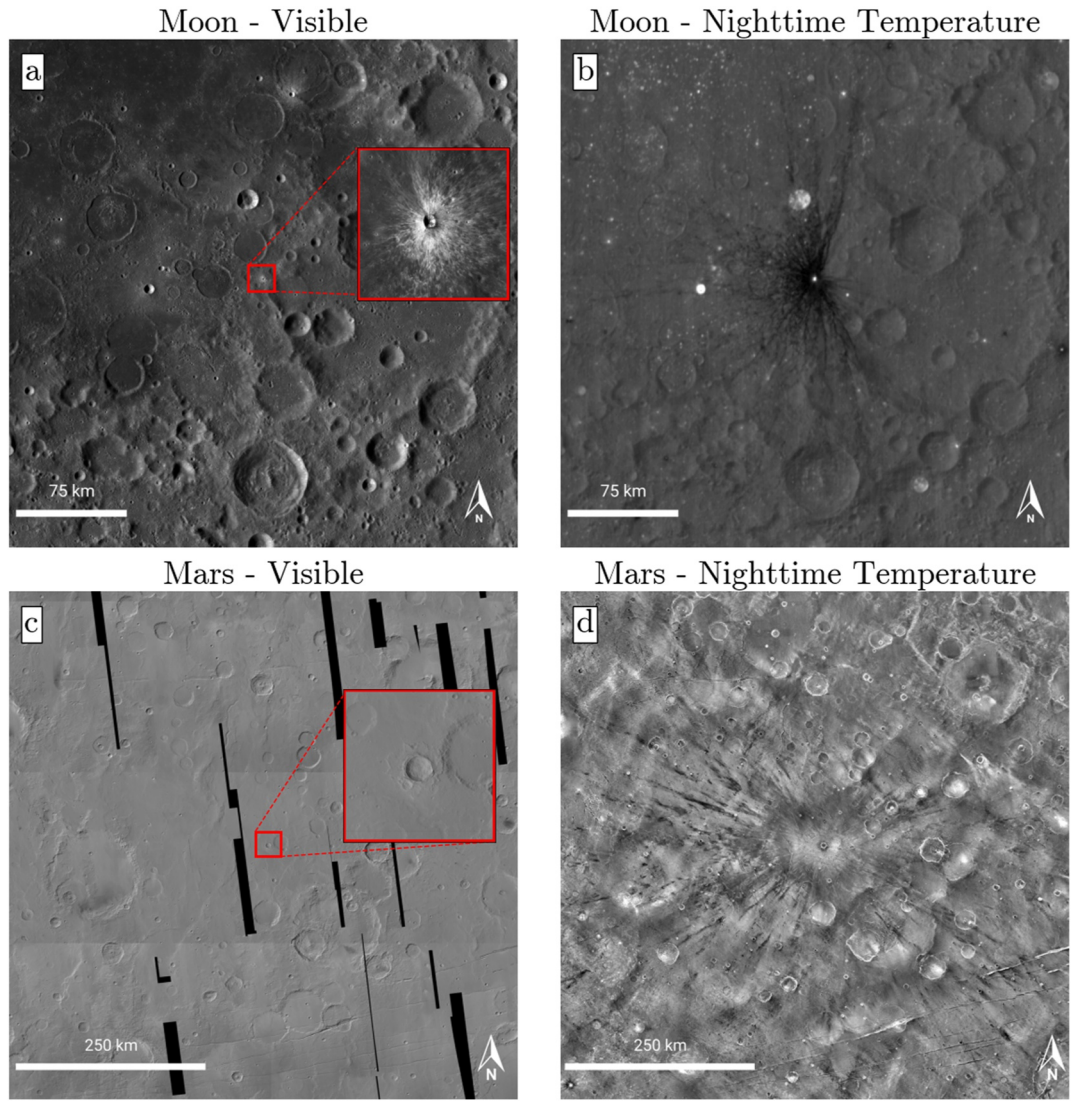


Figure 1. (a) Bandfield crater, a 0.9 km diameter lunar crater located at 5.39°S, 90.76°E. The crater itself is enlarged to show the lengths of the visible crater rays. (b) Nighttime rock-free regolith temperature deviations of the Bandfield crater using LROC WAC global mosaic for shading showing a visible cold spot. Darker portions of the image have a lower surface temperature. (c) Gratteri crater, a 6.9 km diameter Martian crater located at 17.7°S, 199.9°E. As in panel (a), the primary crater is enlarged, but the Martian rays are not visible to the naked eye. (d) Nighttime temperature measurements of Gratteri crater using THEMIS. Darker portions indicate areas with lower temperature reading. Figures adapted from Williams et al. (2018) and Tornabene et al. (2006).

Martian crater rays exhibit notable differences from their lunar counterparts. Mars hosts significantly fewer rayed craters (Tornabene et al., 2006), and its rays are detectable only in thermal infrared wavelengths due to their intrinsically low thermal inertia (see Figure 1d). Unlike the Moon, where impact gardening dominates surface modification, Martian rays are more rapidly eroded by aeolian processes, resulting in a population of comparatively younger rayed craters (Tornabene et al., 2006). Because these rays manifest solely as streaks of low thermal inertia, they are difficult to distinguish in terrains that already possess low thermal inertia. Consequently, Martian rays are primarily observed within young volcanic plains, where the medium to high background thermal inertia enhances their detectability (Mellon et al., 2000; Preblich et al., 2007; Tornabene et al., 2006). Despite these differences, both lunar and Martian rays consist of chains and clusters of secondary craters with overlapping ejecta (Tornabene et al., 2006).

Thermal rays, however, are not unique to Mars. Thermal mapping of the lunar surface using the Lunar Reconnaissance Orbiter (LRO) has revealed similar features surrounding lunar craters known as "cold spots." Analysis

21699100, 2026, 8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JE009609 by Purdue University (West Lafayette), Wiley Online Library on [09/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

21699100, 2026, 8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JE009609 by Purdue University (West Lafayette), Wiley Online Library on [09/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

of LRO Diviner Lunar Radiometer Experiment data by Bandfield et al. (2014) identified these regions as having distinct nighttime temperature drops of up to 10 K. Cold spots exhibit rayed profiles that mimic crater rays, and have lower thermal inertia than their surroundings (Hayne et al., 2017). Figure 1b, adapted from Bandfield et al. (2014), demonstrates the radial symmetry of these cold spots. The formation mechanism of cold spots remains uncertain, though several theories exist, including the idea that an impact can transmit a wave through the ground to "fluff up" the surface material (Frizzell et al., 2025). Recent studies show that all new lunar craters develop cold spots, where their diameter and nighttime temperatures scale directly with impactor size (Powell et al., 2021). Additionally, cold spots extend far beyond the lengths of lunar rays from the same crater (Ghent et al. (2024) and this work).

Rayed craters play a crucial role in remote sensing, helping to establish a chronological framework for planetary geology. While they are not reliable indicators of absolute crater age (Hawke et al., 2004), their presence or absence can be used to determine relative ages of nearby features. Understanding their formation is essential for assessing how they interact with the surrounding terrain, particularly in identifying correlations between the primary crater radius and ray length. Elliott et al. (2018) demonstrated that lunar crater ray length is controlled by the depth of excavation by secondary craters relative to the space-weathered upper regolith layer. This study extends their methodology to Martian craters, which share the same secondary cratering formation process (Tornabene et al., 2006).

In Section 2, we expand on the work done by Elliott et al. (2018) - applying their model of secondary ray formation to Martian crater rays. Section 3 examines lunar cold spots, and the similarities in formation mechanisms and thermal inertia. The minimum excavation depth needed for ray formation on each body is then discussed in Section 4. Finally, we discuss the similarities between Martian crater rays and lunar cold spots and explore potential connections between their formation mechanisms.

2. Constraining Martian Rays

2.1. Crater Ray Modeling

This study adapts the methodology from Elliott et al. (2018) to compare Martian rays and lunar cold spots. Ray mapping in Elliott et al. (2018) was conducted using JMARS (Java Mission-planning and Analysis for Remote Sensing), a geospatial software designed for planetary studies (Dickenshied et al., 2015). Trends observed in the visual data were then explained through secondary crater scaling. Elliott et al. (2018) proposed that following the initial impact, the ejecta curtain launches large boulders that generate secondary craters, and used boulder density mapping around Censorinus crater from Krishna and Kumar (2016) to support this idea. This framework distinguishes three crater ray regimes; Low Energy Deposition, High Energy Deep Excavation, and High Energy Shallow Excavation, described in further detail in Figure 2. According to Elliott et al. (2018), rays are only visible in the first two regimes, as the third regime lacks sufficient energy to excavate the ground at depth of h and expose bright material.

Baldwin (1963) first identified a relationship between primary crater diameter and ray length using photometric data from Kuiper (1960), suggesting that the ray length was proportional to crater diameter. Work by Moore et al. (1974) later demonstrated that a power-law was a better fit for Baldwin's data, yielding a refined relationship for lunar craters defined as:

$$R_r = 5.25R_p^{1.25} \tag{1}$$

where R_r is the length of the crater rays and R_p is the radius of the primary crater.

Elliott et al. (2018) expanded on the work by Moore et al. (1974) by deriving mathematical relationships for the two constants, referring to them as R_1 and s respectively (see Equation 2 below).

The formula derived by Elliott et al. (2018) is:

$$R_r = R_1 R_p^s, \tag{2}$$

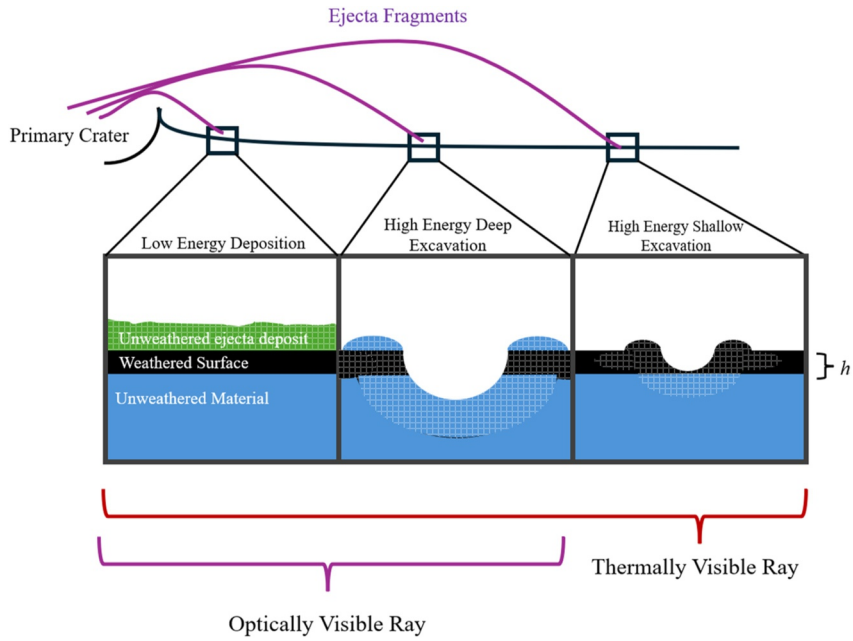


Figure 2. Different crater ejecta regimes proposed by Elliott et al. (2018). The first, low energy deposition, happens closest to the crater rim. It forms when the ejected mass has a low enough velocity that it does not mix with surface material, forming a bright layer of ejecta on top of the pre-existing darkened surface. The second, high energy deep excavation, occurs via secondary cratering, where ejected material has enough mass and velocity to excavate a depth of h through the surface to create visible regions of bright unweathered material. The last regime, high energy shallow excavation, is still caused by secondary cratering, but at distances very far from the crater rim where the ejecta does not have enough mass to excavate a depth of h, so the ray appears to end. Although work by Elliott et al. (2018) defined the ray as only being optically visible in the first two regimes, our work indicates that the ray remains thermally visible across all regimes. Likely caused by a change in the porosity of the material (represented by the hatched regions) around the secondary impactor in the latter two regimes. Figure adapted from Elliott et al. (2018).

where

$$R_1 = \left(\frac{h}{2\alpha}\right)^{\frac{-2-b/(1-\beta)}{a-\beta b/(1-\beta)}} \left|\frac{\ln(1-P)}{-c_2}\right|^{\frac{1}{a-\beta b/(1-\beta)}}, \tag{3}$$

$$s = \frac{a-c}{a-\beta b/(1-\beta)}, \tag{4}$$

and $c_2 = \frac{c_1}{\left(1+\frac{b}{1-\beta}\right)\left(2+\frac{b}{1-\beta}\right)}$ and $c_1 = \pi n_0 d_{\text{boulder}}^{-b} k^{\frac{b}{1-\beta}} \left(\frac{\delta}{\rho}\right)^{\frac{-b}{3(1-\beta)}} R_{ref}^{-c} \left|\frac{b}{1-\beta}\right|$. For the full derivation of these equations, refer to Section 4 of Elliott et al. (2018).

The primary use of the fit derived by Elliott et al. (2018) is to constrain the minimum space-weathered skin depth h needed to form a lunar crater ray (see Figure 2), where a majority of the parameters from Equations 3 and 4 can be constrained from observational data (see Table A1 in the Appendix for a list of these values and their descriptions). In this work, we adopt the same parameter h, but generalize its interpretation: rather than tying it specifically to space weathering, we use h to represent the minimum excavation depth required for ray formation across planetary bodies. This distinction is important because, unlike lunar albedo rays, Martian crater rays and cold spots are not governed by space-weathered skin depth. Thus, while the numerical value of h remains consistent with Elliott et al. (2018), its physical interpretation is broadened to accommodate different surface processes. We also examine the effects of an impact on the thermal inertia of a material. The thermal inertia of a material is dictated by the thermal conductivity, κ, and the density of the material itself, ρ. Equation 5 shows this relationship, with κ in units of $\text{Wm}^{-1}\text{K}^{-1}$, ρ in $\text{kg}\cdot\text{m}^{-3}$, and C, the material's specific heat capacity, is in units of $\text{J kg}^{-1}\text{K}^{-1}$.

21699100, 2026, 8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JE009609 by Purdue University (West Lafayette), Wiley Online Library on [09/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

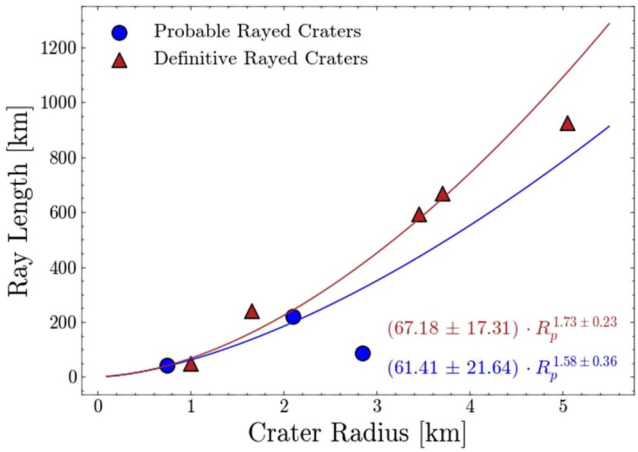


Figure 3. A comparison of the fit lines calculated by including the potential rayed craters described in Tornabene et al. (2006). The red line represents the fit line where only definitive rayed craters were included, and the blue line represents the fit when both definitive and probable rayed craters were included. While the value of R_1 in Equation 2 remains fairly similar, the value of s changes by about 1.1x. Regardless, we chose to rely on the expertise of the researchers in Tornabene et al. (2006) and deferred to their assessment of these probable craters, ultimately removing them from our analysis, and using only four rays from Tornabene et al. (2006) and one ray from McEwen et al. (2005).

$$I = \sqrt{\kappa \rho C} \tag{5}$$

As seen from Equation 5, any reduction in density or thermal conductivity leads to a decrease in the thermal inertia. Thus, we believe that Martian crater rays are formed by an increase in the porosity of the surface regolith immediately following an impact. This increased porosity decreases the density of the substance and lowers the thermal inertia.

2.2. Martian Craters

A table showing the complete list of Martian craters identified by Tornabene et al. (2006) and McEwen et al. (2005) used in this work can be found in Table A3 of the appendix. Additional Martian rayed craters have since been cataloged in Harris et al. (2023), but the length of the crater rays was not recorded, so in our work we only used data from Tornabene et al. (2006) and McEwen et al. (2005). Of the seven craters presented in Tornabene et al. (2006) four had definite crater rays and three had *probable* rays. The inclusion of these probable crater rays was considered in our analysis and can be seen in Figure 3.

Although our Martian rayed crater data set is limited in size, the craters included are among the freshest examples within their size class (as noted by Tornabene et al. (2006)). Freshness is an important consideration because ray systems on Mars are thought to degrade relatively rapidly due to aeolian processes (Tornabene et al., 2006). Expanding the sample to include additional rays would likely introduce older craters whose rays have undergone substantial erosion or surface modification, systematically biasing the measured lengths toward shorter values. While broader mapping efforts would certainly be valuable for future studies, this potential degradation effect represents an additional factor that must be carefully considered when incorporating larger or more heterogeneous data sets.

The addition of the three probable crater rays modifies both the exponent s and the leading constant R_1, in our calculated power-law fit. While the value of R_1 in Equation 2 remains fairly similar, the value of s changes by about 1.1x. Regardless, we chose to rely on the expertise of the researchers in Tornabene et al. (2006) and deferred to their assessment of these probable craters, ultimately removing them from our analysis, and using only four rays from Tornabene et al. (2006) and one ray from McEwen et al. (2005).

2.3. Fit Parameters

To use the model from Elliott et al. (2018), we need constrained the following values: R_{ref}, the radius of our reference crater; d_{boulder}, the minimum diameter of the boulders around the crater; n_0, the density of the boulders on the rim of the reference crater; ρ, the target density; δ, the projectile density, β and κ, both constants that describe the impacted surface; and α, a scaling constant for the excavation depth. A full list of these values and their units is provided in Table A1. Some values, such as β and κ, both constants for dry sand, remained unchanged. The remaining values were obtained from published literature and are discussed in detail below, and their ranges can be seen in Table A2.

2.3.1. Crater Size and Density

Crater S1094b was chosen to be the reference crater used in the Martian analysis. This crater was created recently and recorded by NASA's InSight mission on 24 December 2021 as a seismic event of magnitude >4 (Posiolova et al., 2022). A hypervelocity impact formed crater S1094b, and it has a diameter of 155 m and a depth of 20–23 m (Posiolova et al., 2022). Tusberti et al. (2024) mapped the boulder density around crater S1094b, providing an analog to the lunar study of Censorius crater by Krishna and Kumar (2016). Their work was also used to obtain values for parameters in Table A1, such as the size of the crater (R_{ref} = 77.5 m) and boulder diameters (d_{boulder} = 1.5 m). To refine the analysis, Tusberti categorized boulders into icy and rocky types, but only rocky boulders were considered in this study to align with Elliott et al. (2018). In addition, Tusberti et al. (2024)

21699100, 2026, 8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JE009609 by Purdue University (West Lafayette), Wiley Online Library on [09/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

measured the boulder density twice—first in February 2022 and then again in December 2022, finding different densities each time. February had a higher boulder density, with 2304 boulders/km^2 with $d_{\text{boulder}} \geq 1.50$ m. The density in December was significantly lower, with only 462 boulders/km^2, an 80% loss in counts. Both extremes were used to test fits, however, they did not affect the final outcome, likely due to the weak dependence of R_1 on boulder density (see Equation 3). The results shown in Figure 5 use the February boulder density to more closely match the boulder density of 4327 boulders/km^2 found by Krishna and Kumar (2016) around Censorinus crater on the Moon.

2.3.2. Dependence on Crater Size

Following the procedure given in Elliott et al. (2018), the values of a; ejecta fragment cumulative size frequency distribution (CSFD), b; and the excavation depth, h, were determined. By solving Equations 3, 4 for $R_1 = 67.18$ and $s = 1.73$ respectively, Figure 5 was created to show potential solutions to the fit by Elliott et al. (2018). The intersections of the dashed and solid lines represent points where both Equations 3, 4 are satisfied.

In the fit created by Elliott et al. (2018), the value h represents the skin depth, or the depth of the mature lunar soil. A secondary impact needs to excavate this much soil at a minimum to bring bright material to the surface and contribute to creation of a ray. A value of $h = 10$ cm means that a secondary impact would need to excavate at least 10 cm of regolith to form an optically visible ray (this is known as High Energy Deep Excavation, see Figure 2). A secondary impactor that can only excavate 9 cm of material in that region would fall into the High Energy Shallow Excavation regime, and an optical ray would not be visible as the impactor would penetrate through the dark space-weathered surface material. However, the model by Elliott et al. (2018) uses the h parameter to only consider optical rays, and not thermal rays, which may form with excavation depths less than h. Since the surface regolith on Mars lacks a space-weathered surface, we consider h the minimum depth of excavation needed to form a ray, and refer to it as the excavation depth (see Section 2.1). We apply this definition to both thermal and optical rays in this work.

Although larger values of h were tested, the values shown in Figure 5b are the excavation depth values that intersected with the c curves. Martian values of h are only slightly smaller than those calculated by Elliott et al. (2018) for lunar soil (mm to cm scale on Mars compared to ∼5–10 cm on the Moon). The surface of Mars experiences aeolian weathering, and if this was the true excavation depth, then the winds on Mars would overturn the young soil almost continuously. However, we believe that the excavation depth value may not represent the depth of material disruption, and we discuss this further in Section 5.

The values of both c and P are difficult to approximate, but have extremes that are easier to visualize. The dependence on primary crater size is represented by c. A value of $c = 0$ would mean that the number of boulders would not depend on the size of the primary crater, and a value of $c = 1$ would mean that the boulder population of the crater is linearly dependent on the size of the boulder. Values of c ranging from 0 to 1 were tested and shown in Figure 5. The value P represents the probability (from 0 to 1) that a crater will excavate the surface to a depth h. In a scenario where $P = 0$, there would be no overturn of the surface material, and no rays would form, since Equation 2 would equal zero. By contrast, a value of $P = 1$ would mean that every piece of ejecta would excavate the surface to a depth of h and be visible as bright unweathered secondary ejecta. Elliott et al. (2018) assumes a value of $P = 0.5$, but this work varied P between 0.2 and 0.8 to ensure a more robust coverage. Results with $P = 0.5$ are shown in Figure 5.

A comparison of the crater radii to ray lengths of Martian and lunar crater rays using Equation 2 can be seen in Figure 4.

3. Comparing Martian Rays and Lunar Cold Spots

Despite having different proposed formation mechanisms, the physical properties of lunar cold spots and Martian crater rays are very similar. Both have radial ejecta patterns with low thermal inertia (Hayne et al., 2017; McEwen et al., 2005), ejecta patterns visible with thermal imaging (Bandfield et al., 2014; Preblich et al., 2007), and are expected to be a few centimeters thick (Bandfield et al., 2014; Hayne et al., 2017; Tornabene et al., 2006). One major difference they have is their estimated lifetimes. Lunar cold spots are only expected to last a few hundred thousand years (Williams et al., 2018), whereas Martian crater rays have lifetimes on the order of tens of Mya (Harris et al., 2023; Mc Keown et al., 2019). This is most likely explained by the impact gardening rate on the

21699100, 2026, 8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JE009609 by Purdue University (West Lafayette), Wiley Online Library on [09/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

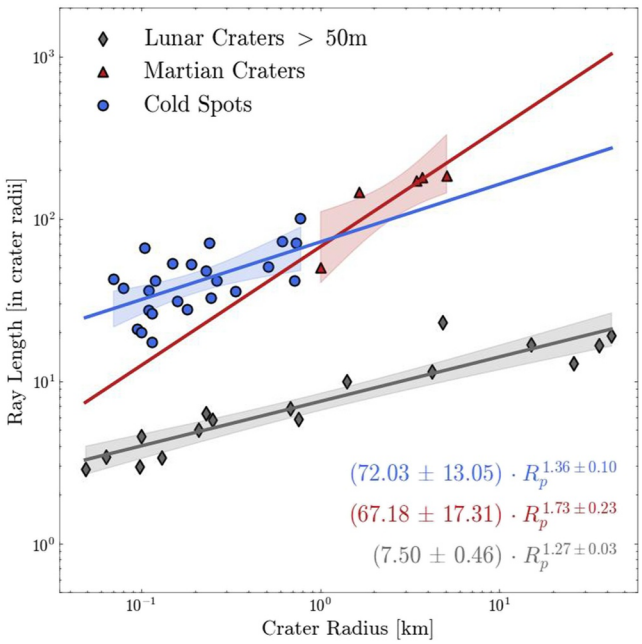


Figure 4. A comparison of crater ray length and the crater radius calculated for visible albedo lunar rays by Elliott et al. (2018) (grey), Martian crater rays from McEwen et al. (2005) and Tornabene et al. (2006) (red), and lunar cold spots from Bandfield et al. (2014) (blue). Although many more rayed and cold spot craters have been detected, only a small fraction have had the length of their rays determined. Confidence intervals at 95% confidence are shown for all three lines. The power-law index values for cold spots and albedo lunar rays are nearly identical, differing only slightly outside their respective error ranges. However, the leading constant, R_1, differs by more than an order of magnitude, with both Martian crater rays and cold spots exhibiting substantially larger values than albedo lunar rays. The leading constant for Martian rays falls within the error range of the value for lunar cold spots. Lunar craters from Elliott et al. (2018) with diameters smaller than 50 m were not used in this study. Some of these small craters have rays longer than predicted by the Elliott model and may represent bright ejecta from the primary crater deposited at the surface. Craters of similar sizes were not considered for Martian rays or lunar cold spots (see Section 3 for additional information).

Moon being significantly larger than that on Mars (Melosh, 1989), which would allow surface features such as rays to exist for longer stretches of geological time on Mars.

In this work, we use cold spot data from Bandfield et al. (2014), Martian crater ray data from Tornabene et al. (2006) and McEwen et al. (2005), and lunar crater data from Elliott et al. (2018). We acknowledge that our selection of cold spots and rayed Martian craters is relatively small, but out of the many more rayed and cold spot craters have been detected, only a small fraction have had the length of their rays determined. The focus of this work is on the comparison of rayed features on different bodies, and as a result, only values from the above publications were used in our analysis.

Additionally, Elliott et al. (2018) examined the behavior of rays around small craters (<50 m in diameter) and found that they had rays longer than predicted by their model. They offered the explanation that these points are outliers produced primarily by deposition of low velocity ejecta from the primary crater itself rather than excavation of bright material by secondary impacts. These small craters are not expected to produce secondary craters at these distances. Section 4.3 of Elliott et al. (2018) explains this in further detail. This explanation is further supported by Williams et al. (2018), who found that the crater size-frequency distributions of smaller diameter craters exhibit a much shallower slope compared to medium or larger craters. They attributed this either to the difficulty of accurately measuring the lengths of small cold spots or to additional factors influencing cold spot formation at lower impact energies. Since impact craters of comparable sizes are not present in our Martian or cold spots data sets, we excluded these small lunar craters from our analysis to prevent this end member behavior from disproportionately influencing only one data set and biasing our cross comparison.

Analysis using the Elliott model of ray formation was performed on lunar cold spots as well, and compared to both lunar and Martian rays in Figure 4. Despite having different values of s and R_1, the majority of cold spot data points align with the Martian ray data. This may suggest that the true formation mechanism of both lunar cold spots and Martian crater rays are linked.

4. Excavation Depth and Depth of Disruption

To calculate the value of the excavation depth, Equation 2 was solved for the values of a, b, and h, by plugging in the calculated values of s and R_1 shown in Figure 4 for each fit. The results of this parameterization can be seen in Figure 5.

Elliott et al. (2018) derived a value for the excavation depth of lunar rays on the order of 10 cm, which agrees with prior literature (see Bandfield et al. (2014); Hayne et al. (2017)). However, applying this same model to lunar cold spots and Martian crater rays shows a much smaller excavation depth (on the order of 1 mm–5 cm). While these excavation depths may seem low compared to albedo rays, it matches the expectation we would have for thermal inertia rays (see Figure 2). As discussed in the following section, impact disruption can extend 10–100 times deeper than the depth of excavation.

Recent data from Tusberti et al. (2024) shows that the boulder density around the Martian crater S1094b is $b = -4.68$ for the February distribution and $b = -3.47$ for the December distribution, both of which correspond to small excavation depth values (h). Similar work by Button (2017) on boulder distribution in the Tharsis Region and Elysium Planitia revealed a comparable density profile, with values of b in the same regime ($b \approx -3$).

To further investigate this, we tested boulder parameters from Elliott et al. (2018) on Martian crater data and found that they only had a modest effect on the results (see Figure A1 in the Appendix). This suggests that the

21699100, 2026, 8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JE009609 by Purdue University (West Lafayette), Wiley Online Library on [09/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

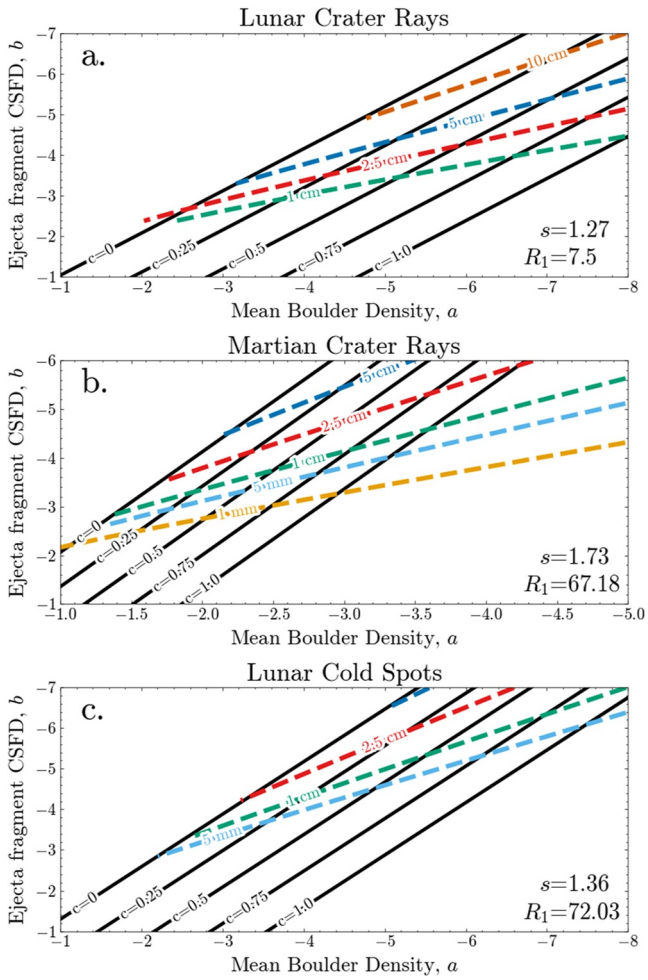


Figure 5. Plots of the minimum excavation depth needed to form a crater ray for different geologic features using the methodology in Elliott et al. (2018). The solid black lines are contours of c that satisfy the value of s written in the lower right. The dashed lines represent different values of h, the excavation depth, for the value of R_1 for each scenario. Intersection points of h and the c contours are values at which all components of the empirical fit derived by Elliott et al. (2018) are satisfied. Note that x and y axes change in scale for Figure 5b. (a) A recreation of Figure A2 from Elliott et al. (2018), using the lunar parameters from Table A1 derived by Elliott. (b) Excavation depth of the Martian crater rays from Tornabene et al. (2006) and McEwen et al. (2005), using the Martian density parameters derived from literature (see Table A2). (c) Excavation depth of the lunar cold spots from Bandfield et al. (2014) using the lunar parameters derived by Elliott et al. (2018).

observed differences between lunar and Martian rays are not solely due to variations in boulder distributions, but rather point to a difference in the underlying physics behind the different types of ray creation. While Elliott's framework accurately predicts the length and skin depth of visible albedo rays, thermal rays may exhibit distinct properties not accounted for in their model.

5. Discussion

The goal of this work was to compare the lengths of Martian rayed craters to those found on the Moon. Given the large difference in lunar and Martian R_1 values shown in Figure 4, it seems that the function derived by Elliott et al. (2018) may be missing some fundamental physics behind ray formation. The values of both the power law index, s, and the initial constant, R_1, increase in both the Martian and cold spot scenario, with the calculated Martian R_1 value being nearly 10x the size of the lunar R_1 given by Elliott. This is apparent in Figure 4 where lunar rays are depicted alongside Martian rays. A rayed crater with the same radius on both bodies would have rays nearly 10x longer on Mars compared to the Moon.

To determine if our chosen crater, S1094b, affected our results, we also tested the Elliott model on the Martian crater diameters and ray lengths using the lunar parameters that Elliott et al. (2018) derived. This means we used a lunar crustal density, projectile density, and Censorinus crater on the Moon as the reference crater. Despite all these changes, we found that the excavation depth contours did not change by a significant amount. Further detail can be found in the Appendix.

Longer ray lengths on Mars initially seems to go against intuition, as Mars has a higher gravity than the Moon, and the presence of a thin atmosphere. However, as shown by Elliott et al. (2018), when assuming ballistic trajectories for crater rays, the gravity term itself cancels out and would not affect ray formation (see Equations 4–6 in Elliott et al. (2018)). Mars' thin atmosphere is also thought to have little effect on objects larger than 20 cm (Artemieva & Ivanov, 2004), which is smaller than many secondary impactors.

In this work, lunar cold spots have been compared to both Martian crater rays and visible albedo lunar rays. Lunar cold spots and Martian rays have been shown to have similar features, including low thermal inertia (Hayne et al., 2017; Preblich et al., 2007) and, as this work found, both lunar cold spots and Martian thermal rays have lengths an order of magnitude longer than visible optical rays found around lunar craters (see Figure 4). To further investigate this connection, we now examine how these features may share a common formation process.

Recent work by Ghent et al. (2024) discussed thermophysical rays on the Moon, which are warm anomalies that form around a rayed crater near the rim. Unlike cold spots and low thermal inertia spots, the surfaces of these thermophysical rays have *increased* temperatures compared to the surrounding material. However, this may not be entirely unique to the Moon. Tornabene et al. (2006) discusses similar warm anomalies with high thermal inertia around Martian rayed craters, and these thermophysical rays can even be seen around lunar cold spot craters (such as Bandfield crater, see Figure 1) A depiction of this can be seen in Figure 6. Warm thermophysical rays such as those included in Harris et al. (2023) likely last longer than cold rays such as those found in Tornabene et al. (2006). To our knowledge, no prior publication has quantified the lengths of warm lunar thermophysical rays. As such, the ejecta behavior we propose should be viewed as a theoretical model intended to guide future observational work.

21699100, 2026, 8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JE009609 by Purdue University (West Lafayette), Wiley Online Library on [09/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

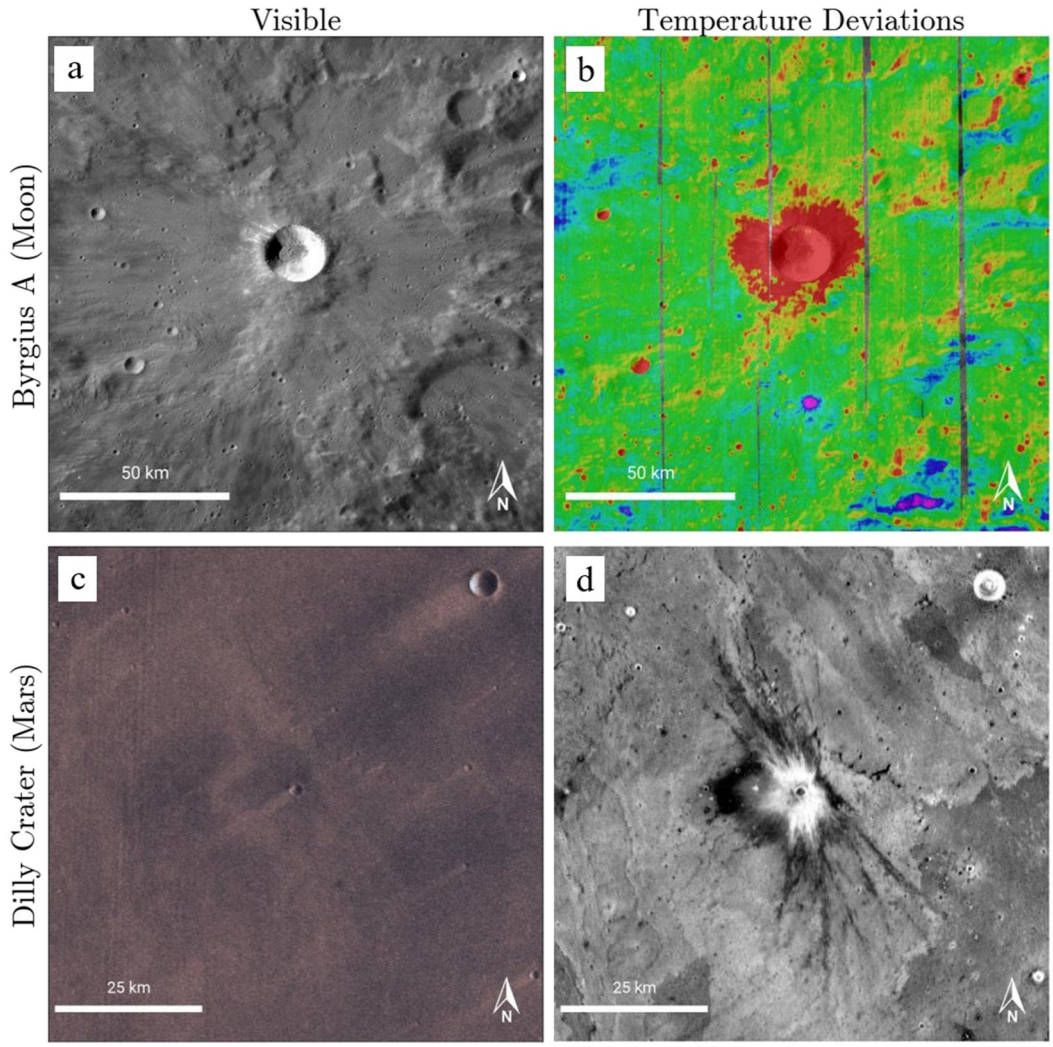


Figure 6. A comparison of thermophysical rays on the Moon and Mars. (a) Byrgius A, a lunar crater located at 24.55°S, 296.2°E with a diameter of 18.2 km. (b) LRO Diviner Normalized Rock Free Regolith Temperature of Byrgius A showing warm, thermophysical rays (red) in and around the crater rim. (c) Dilly, a Martian crater located at 13.266°N, 157.22°E. (d) Nighttime temperature of Dilly crater using THEMIS, showing warm thermophysical rays (white) and colder thermal inertia rays (black).

In our model, secondary cratering serves as the primary mechanism for the disruption of the surface regolith at long distances, contributing to the formation of both Martian crater rays and lunar cold spots. Figure 5 shows that excavation depths for Martian crater rays and lunar cold spots are significantly smaller than those of albedo-based lunar rays. This fits with our understanding following the logic presented in Figure 2. Prior work (Wiggins et al. (2019, 2022)) has shown that hypervelocity impactors generate rarefaction waves, inducing porosity changes in materials up to 1–10 crater diameters away. Using the relationship between excavation depth and crater diameter from Melosh (1989), these porosity changes would occur at 10–100 times the excavation depth. While Wiggins et al. (2019, 2022) focused on primary hypervelocity impacts (10–20 km/s), the resulting disruption from the rarefaction wave extends far beyond the crater itself. This effect would likely persist in secondary impacts, which occur at much lower velocities (<1 km/s) (Melosh, 1989).

We can do a very conservative estimate using Martian crater rays. One of the calculated minimum excavation depth values was 1 cm. If we assume that this crater was formed via a secondary impact, and that it only affected surrounding material 2 crater radii away, that means that the depth of material disruption would be 20 times our original excavation depth, or a 20 cm area around the impact. This disrupted area would have a higher porosity than the surrounding material, and by Equation 5, a lower thermal inertia too.

21699100, 2026, 8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JE009609 by Purdue University (West Lafayette), Wiley Online Library on [09/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

21699100, 2026, 8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JE009609 by Purdue University (West Lafayette), Wiley Online Library on [09/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Martian crater rays are often found in regions of low to moderate albedo (0.10–0.26) and high background thermal inertia (110–330 J m^{-2} K^{-1} $s^{-1/2}$), defined as Thermophysical unit C by Mellon et al. (2000) and Putzig et al. (2005). This region is believed to be duricrust, a formation consisting of hardened and solidified surface regolith (Putzig et al., 2005). When an impactor strikes a duricrust-covered portion of the Martian surface, the impact would produce a rarefaction wave that breaks up the duricrust, causing the porosity to increase and the density to decrease (Wiggins et al., 2019, 2022). By Equation 5, this effect would cause a decrease in the thermal inertia, leading to the ray-like patterns of low thermal inertia seen on the Martian surface. While a majority of rayed craters are found within Unit C, a smaller subset is found within Unit B (Harris et al., 2023), which, like Unit C, exhibits high background thermal inertia (160–355 J m^{-2} K^{-1} $s^{-1/2}$, Putzig et al. (2005)). This thermal contrast between the rays and their surroundings enables the low-inertia streaks to remain thermally distinct and thus detectable in orbital data.

On the Moon, surface regolith is often not as compact as Martian duricrust. However, micrometeorite bombardment has been known to cause small impact melts on the lunar surface (Keller & McKay, 1997; Stojic et al., 2023; Wu et al., 2017). If these micrometeorite impacts were to be sufficient enough to create a thin layer of weak cementation over a period of geologic time, it would slowly increase the thermal inertia and conductivity. The rarefaction wave created by a large impactor would then be able to disrupt or modify this surface layer, reducing the thermal conductivity, which in turn would lower the surface's thermal inertia, creating the lunar cold spots we see today. We speculate that the formation of impact glass and agglutinates may increase thermal conductivity, and that subsequent disruption of this material may reduce it. It is also possible the typical state of the regolith is slightly compacted compared to newly disrupted regolith. Frizzell et al. (2025) hypothesize that impacts produce a surface dilating wave that can create the observed decompaction associated with cold spots. Their model indicates that the radial extent of this wave and how it corresponds to cold spot size depends on the layered structure of the Moon's shallow subsurface. It is unclear if this model could be extended to Martian rays, as many of these rays are found on young volcanic plains which have a more consolidated subsurface structure than the Moon. Thus, the work of Frizzell et al. (2025) may suggest that Martian rays have a distinct origin from lunar cold spots.

If what we propose is the true mechanism of thermal ray formation, then the High Energy Shallow Excavation regime depicted in Figure 2 represents the cutoff point for visible rays. The thermal rays visible on both the Moon and Mars would thus be just an extension of the visible rays we already can see.

6. Conclusions

To summarize the points made in the discussion section, this work has done the following:

1. Shown that Martian crater rays are longer than lunar ones by an order of magnitude for craters with similar diameters (see Figure 4).
2. Demonstrated that Martian craters have ray lengths similar to those found around lunar cold spots (see Figure 4).
3. Proposed a relationship between the formation mechanisms of lunar cold spots and Martian crater rays
4. Expanded upon the ray formation model by Elliott et al. (2018) by including thermal rays in Figure 2.
5. Proposed that Martian crater rays and lunar cold spots may be extensions of optically visible rays.

Future work should further investigate the relationship between lunar cold spots and Martian crater rays. This work only used a small subset of Martian craters and lunar cold spots with publicly available ray lengths, but there are other cataloged rayed craters and cold spots that do not have recorded ray lengths (see Harris et al. (2023)). Including more Martian rayed craters and lunar cold spots may show a further correlation between these two geologic features. This also applies to warm, thermophysical rays. Future analysis of the lengths of these warm thermophysical rays could provide further insight into ejecta-surface interactions. Additionally, crater rays are not only exclusive to Mars and the Moon. Mercury also has a number of crater rays with similar physical properties to those on Mars. Incorporating Mercurian rays in a future study may provide further insights into thermal ray formation.

Updates to models such as the one by Elliott et al. (2018) should be considered to more accurately predict the lengths of thermally visible crater rays. The point where rays are no longer optically visible may not mark the end of their formation, but rather the limit of their detectability by the naked eye.

Appendix A

The appendix below has three tables and two supplementary figures. Tables A1 and A2 describe the parameters used in the Elliott et al. (2018) fit, including values used, their units, and source references. Table A3 lists the location and radius of the Martian craters used in this study. Figures A1 and A2 are variants of figures that are already present in the main paper. Figure A1 asks what would happen if Martian parameters were not used for the analysis of Martian crater rays in Figure 5 and Figure A2 is a recreation of Figure 4, but with the *y*-axis scaled to crater ray length in km instead of crater radii.

Table A1
A Description of the Parameters Used in Elliott et al. (2018)

Parameter	Units	Description
a	Unitless	Mean Spatial Density of Boulders
b	Unitless	Ejecta Fragment Cumulative Size Frequency Distribution
c	Unitless	Dependence on Crater Size
P	Unitless	Probability a surface is excavated to a depth h
β	Unitless	Constant 0.17 for dry sand (Schmidt & Housen, 1987)
k	Unitless	Empirical Scaling Constant 0.625 for Dry Sand (Schmidt & Housen, 1987)
α	Unitless	Maximum excavation depth of a streamline (Gault et al., 1974)
h	km	Depth of Excavation
R_{ref}	km	Radius of the reference crater
$d_{boulder}$	km	Minimum Diameter of boulders around the Crater
n_0	Count per km^2	Boulder Density at the Rim of the Reference Crater
δ	kg/m^3	Projectile Density
ρ	kg/m^3	Target Density

Note. Most values are unitless, but those with distance are all in km instead of m. Note: This is a necessary part of the fit derived by (Elliott et al., 2018). The function they derived in Equations 3 and 4 only works with distance units in kilometers, not meters.

Table A2
An Overview of the Values Tested While Fitting the Ray Scaling Law Function Given in Elliott et al. (2018) to the Martian Craters From McEwenm et al. (2005) and Tornabene et al. (2006)

Parameter	Values tested	Source
a	−2 to −9	Obtained from Fit
b	−2 to −9	Obtained from Fit
c	0 to 1	Elliott et al. (2018)
P	0.2 to 0.8	Elliott et al. (2018)
h	100 μm to 1 m	Obtained from Fit
R_{ref}	77.5 m	Tusberti et al. (2024)
$d_{boulder}$	1.5 m	Tusberti et al. (2024)
n_0	462 to 2308 boulders/km^2	Tusberti et al. (2024)
δ	3,100 kg/m^3	Baratoux et al. (2014)
ρ	2,582 kg/m^3	Goossens et al. (2017)

Note. The surface density of 2582 kg/m^3 is the crustal surface density of Mars, and the projectile density of 3,100 kg/m^3 is the estimated density of surface boulders.

21699100, 2026, 8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JE009609 by Purdue University (West Lafayette), Wiley Online Library on [09/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table A3
This Table Shows the Martian Craters Used in This Study

Crater name	Radius [km]	Lat.	Long.	Median ray length [km]	Source
Zunil	5.05	7.70	166.00	927	McEwen et al. (2005)
Tomini	3.70	16.27	125.90	668	Tornabene et al. (2006)
Gratteri	3.45	−17.70	199.9	595	Tornabene et al. (2006)
Zumba	1.65	−28.65	226.90	240	Tornabene et al. (2006)
Dilly	1.00	13.27	157.23	50	Tornabene et al. (2006)
*Tomini B**	2.10	14.90	123.25	220	Tornabene et al. (2006)
*Crater A**	2.85	18.10	155.50	86	Tornabene et al. (2006)
*Crater B**	0.75	15.50	159.20	42	Tornabene et al. (2006)

Note. The Data on the craters themselves are taken from Tornabene et al. (2006), except for crater Zunil, which is taken from McEwen et al. (2005). A Total of Eight Rayed Craters can be Found on Mars, but only the first five were Used in this study. The last three craters, Tomini B, Crater A, and Crater B, were determined as only *probable* rayed craters by Tornabene et al. (2006), so they were not included in this analysis.

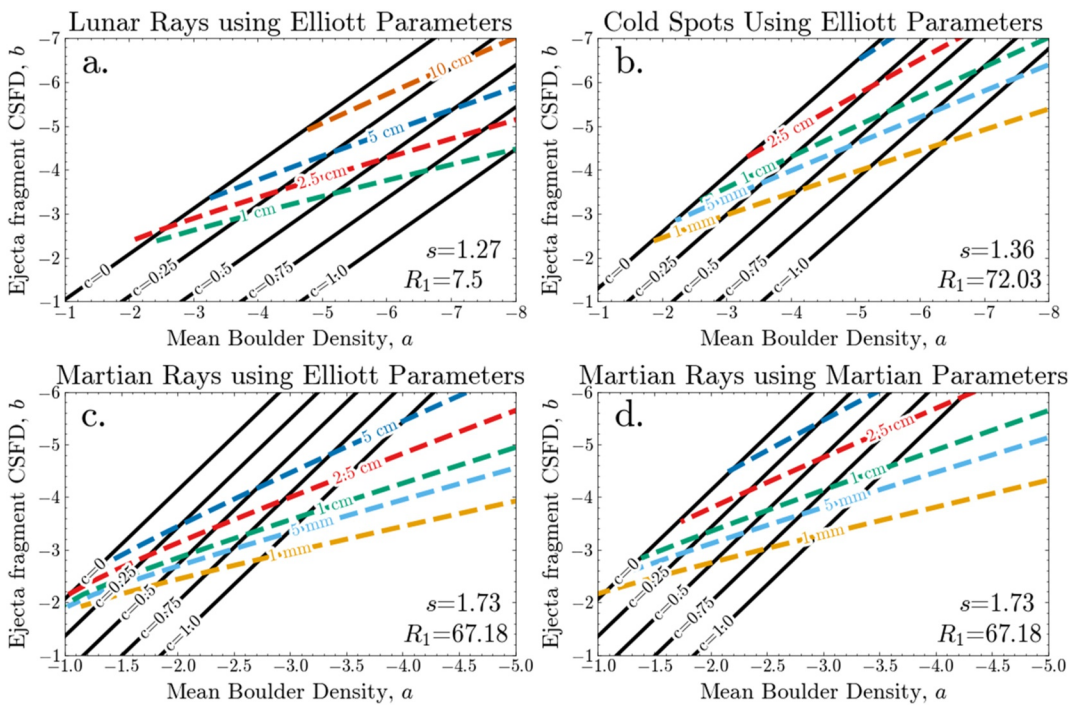


Figure A1. A comparison of lunar rays, cold spots, and Martian rays, both with Lunar parameters from Elliott et al. (2018) and Martian parameters shown in Table A2. It should be noted that Panels (a, b and d) correspond to the plots in Figures 5a–5c respectively. (a) A recreation of Figure A2 from Elliott et al. (2018) using the lunar parameters from Table A1 derived by Elliott. These values are specific to the Moon, and include crustal density, projectile density, reference crater size, and boulders around the reference crater. (b) Excavation depth of the lunar cold spots from Bandfield et al. (2014) using the lunar parameters derived from Elliott et al. (2018) (c) Excavation depth of the Martian crater rays from Tornabene et al. (2006) and McEwen et al. (2005), using the lunar parameters derived by Elliott. (d) Excavation depth of the Martian crater rays from Tornabene et al. (2006) and McEwen et al. (2005), using the Martian density parameters derived from literature (see Table A2). Statistical *t*-tests were performed comparing the skin depth contours of panels (c and d) to determine if the effect seen was a byproduct of the specific crater chosen in this work. Skin depths of 5 cm and greater were statistically significant, but the smaller skin depths (of 2.5 cm and below) were determined to be not statistically different.

21699100, 2026, 8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JE009609 by Purdue University (West Lafayette), Wiley Online Library on [09/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

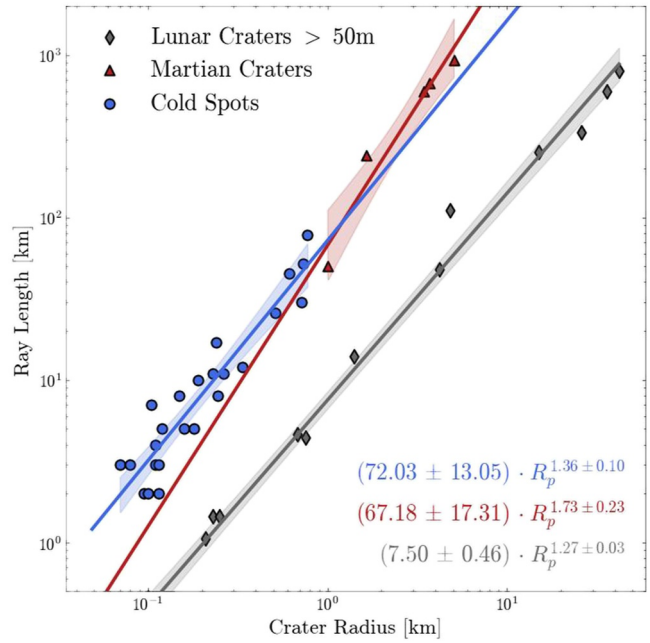


Figure A2. Note: This is a recreation of Figure 4, but with the y-axis scaled to crater ray length in km instead of crater radii. A comparison of crater ray length and the crater radius calculated for visible albedo lunar rays by Elliott et al. (2018) (grey), Martian crater rays from McEwen et al. (2005) and Tornabene et al. (2006) (red), and lunar cold spots from Bandfield et al. (2014) (blue). Lunar craters from Elliott et al. (2018) with diameters smaller than 50 m were not used in this study. Some of these small craters have rays longer than predicted by the Elliott model and may represent bright ejecta from the primary crater deposited at the surface. Craters of similar sizes were not considered for Martian rays or lunar cold spots (see Section 3 for additional information).

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

In this work, data from a number of prior papers was used for all fits. This includes lunar crater data from Table A2 of Elliott et al. (2018), Martian crater ray data from Table A1 of Tornabene et al. (2006) (also as Table A3 in this paper), and lunar cold spot data from Table A1 of Bandfield et al. (2014). Figures were made with Matplotlib version 3.10.0 (Caswell et al., 2020; Hunter, 2007), available under the Matplotlib license at https://matplotlib.org/.

Acknowledgments
This work was supported by the NASA SSERVI Lunar Structure, Composition, Processes and Evolution (LunaSCOPE) Grant 80NSSC23M0161. We would like to thank both of our reviewers for their insightful comments that helped us to improve the manuscript. Finally, special thanks go to Mia Rudin for her steadfast support and encouragement throughout this process.

References

Artemieva, N., & Ivanov, B. (2004). Launch of martian meteorites in oblique impacts. *Icarus*, *171*(1), 84–101. https://doi.org/10.1016/j.icarus.2004.05.003

Baldwin, R. B. (1963). The measure of the moon. In *Books on demand*.

Bandfield, J. L., Song, E., Hayne, P. O., Brand, B. D., Ghent, R. R., Vasavada, A. R., & Paige, D. A. (2014). Lunar cold spots: Granular flow features and extensive insulating materials surrounding young craters. *Icarus*, *231*, 221–231. https://doi.org/10.1016/j.icarus.2013.12.017

Baratoux, D., Samuel, H., Michaut, C., Toplis, M. J., Monnereau, M., Wieczorek, M., et al. (2014). Petrological constraints on the density of the martian crust. *Journal of Geophysical Research: Planets*, *119*(7), 1707–1727. https://doi.org/10.1002/2014JE004642

Button, N. (2017). *Boulder distribution analysis of impact craters in the tharsis Region and elysium planitia*. LSU Master's Theses.

Caswell, T. A., Droettboom, M., Lee, A., Hunter, J., Firing, E., Sales De Andrade, E., et al. (2020). Matplotlib/matplotlib: Rel: V3.3.0. Zenodo. https://doi.org/10.5281/zenodo.3948793

Dickenshied, S., Anwar, S., Noss, D., Hagee, W., & Carter, S. (2015). JMARS — Easy visualization and analysis of planetary remote sensing data. In *LPI editorial board.* In *Second planetary data workshop* (Vol. 1846).7013.

Elliott, J. R., Huang, Y.-H., Minton, D. A., & Freed, A. M. (2018). The length of lunar crater rays explained using secondary crater scaling. *Icarus*, *312*, 231–246. https://doi.org/10.1016/j.icarus.2018.04.015

Frizzell, E. S., Hartzell, C. M., & Schmerr, N. C. (2025). Wave transmission through the megaregolith as a mechanism for lunar cold spot formation. *Icarus*, *429*, 116461. https://doi.org/10.1016/j.icarus.2025.116461

21699100, 2026, 8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JE009609 by Purdue University (West Lafayette), Wiley Online Library on [09/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Gault, D. E., Hoerz, F., Brownlee, D. E., & Hartung, J. B. (1974). Mixing of the lunar regolith. *Lunar and Planetary Science Conference Proceedings*, *3*, 2365–2386.
Ghent, R. R., Costello, E. S., & Parker, A. H. (2024). The population of young craters on the moon: New catalog and spatial and temporal analysis. *The Planetary Science Journal*, *5*(4), 89. https://doi.org/10.3847/PSJ/ad293d
Goossens, S., Sabaka, T. J., Genova, A., Mazarico, E., Nicholas, J. B., & Neumann, G. A. (2017). Evidence for a low bulk crustal density for mars from gravity and topography. *Geophysical Research Letters*, *44*(15), 7686–7694. https://doi.org/10.1002/2017GL074172
Harris, J. K., McKeown, L. E., Parenti, C., Tornabene, L. L., & Grindrod, P. M. (2023). An updated catalog of rayed craters on mars. *Earth and Space Science*, *10*(1), e2021EA001814. https://doi.org/10.1029/2021EA001814
Hawke, B. R., Blewett, D. T., Lucey, P. G., Smith, G. A., Bell, J. F., Campbell, B. A., & Robinson, M. S. (2004). The origin of lunar crater rays. *Icarus*, *170*(1), 1–16. https://doi.org/10.1016/j.icarus.2004.02.013
Hayne, P. O., Bandfield, J. L., Siegler, M. A., Vasavada, A. R., Ghent, R. R., Williams, J.-P., et al. (2017). Global regolith thermophysical properties of the moon from the diviner lunar radiometer experiment. *Journal of Geophysical Research: Planets*, *122*(12), 2371–2400. https://doi.org/10.1002/2017JE005387
Hunter, J. D. (2007). Matplotlib: A 2d graphics environment. *Computing in Science & Engineering*, *9*(3), 90–95. https://doi.org/10.1109/MCSE.2007.55
Keller, L. P., & McKay, D. S. (1997). The nature and origin of rims on lunar soil grains. *Geochimica et Cosmochimica Acta*, *61*(11), 2331–2341. https://doi.org/10.1016/S0016-7037(97)00085-9
Krishna, N., & Kumar, P. S. (2016). Impact spallation processes on the moon: A case study from the size and shape analysis of ejecta boulders and secondary craters of censorinus crater. *Icarus*, *264*, 274–299. https://doi.org/10.1016/j.icarus.2015.09.033
Kuiper, G. P. (1960). *Photographic lunar atlas*. University of Chicago Press.
McEwen, A. S., Preblich, B. S., Turtle, E. P., Artemieva, N. A., Golombek, M. P., Hurst, M., et al. (2005). The rayed crater zunil and interpretations of small impact craters on mars. *Icarus*, *176*(2), 351–381. https://doi.org/10.1016/j.icarus.2005.02.009
Mc Keown, L., Grindrod, P. M., Harris, J. K., & Parenti, C. (2019). An updated global catalogue of rayed craters on Mars: Potential sources for Martian meteorites. *Agu fall meeting abstracts*, *2019*, EP53F–2240.
Mellon, M. T., Jakosky, B. M., Kieffer, H. H., & Christensen, P. R. (2000). High-resolution thermal inertia mapping from the mars global surveyor thermal emission spectrometer. *Icarus*, *148*(2), 437–455. https://doi.org/10.1006/icar.2000.6503
Melosh, H. (1984). Impact ejection, spallation, and the origin of meteorites. *Icarus*, *59*(2), 234–260. https://doi.org/10.1016/0019-1035(84)90026-5
Melosh, J. (1989). *Impact cratering*. Oxford University Press.
Moore, H. J., Hodges, C. A., & Scott, D. H. (1974). Multiringed basins - Illustrated by Orientale and associated features. *Lunar and Planetary Science Conference Proceedings*, *1*, 71–100.
Neish, C. D., Blewett, D. T., Harmon, J. K., Coman, E. I., Cahill, J. T. S., & Ernst, C. M. (2013). A comparison of rayed craters on the moon and mercury. *Journal of Geophysical Research: Planets*, *118*(10), 2247–2261. https://doi.org/10.1002/jgre.20166
Posiolova, L. V., Lognonné, P., Banerdt, W. B., Clinton, J., Collins, G. S., Kawamura, T., et al. (2022). Largest recent impact craters on mars: Orbital imaging and surface seismic co-investigation. *Science*, *378*(6618), 412–417. https://doi.org/10.1126/science.abq7704
Powell, T., Greenhagen, B., Taylor, S., Speyerer, E., Williams, J.-P., Hayne, P., & Paige, D. (2021). New lunar cold spots: The thermophysical properties of the youngest craters on the Moon. *Agu fall meeting abstracts*, *2021*, EP24B-01.
Preblich, B. S., McEwen, A. S., & Studer, D. M. (2007). Mapping rays and secondary craters from the martian crater zunil. *Journal of Geophysical Research*, *112*(E5). https://doi.org/10.1029/2006JE002817
Putzig, N. E., Mellon, M. T., Kretke, K. A., & Arvidson, R. E. (2005). Global thermal inertia and surface properties of mars from the mgs mapping mission. *Icarus*, *173*(2), 325–341. https://doi.org/10.1016/j.icarus.2004.08.017
Schenk, P. M., & McKinnon, W. B. (1991). Dark-ray and dark-floor craters on Ganymede, and the provenance of large impactors in the Jovian system. *Icarus*, *89*(2), 318–346. https://doi.org/10.1016/0019-1035(91)90181-R
Schmidt, R. M., & Housen, K. R. (1987). Some recent advances in the scaling of impact and explosion cratering. *International Journal of Impact Engineering*, *5*(1), 543–560. Hypervelocity Impact Proceedings of the 1986 Symposium. https://doi.org/10.1016/0734-743X(87)90069-8
Stojic, A. N., Weber, I., Morlok, A., Pavlov, S. G., Hiesinger, H., Reitze, M. P., et al. (2023). Simulation of surface regolith gardening and impact associated melt layer production under ns-pulsed laser ablation. *Icarus*, *391*, 115344. https://doi.org/10.1016/j.icarus.2022.115344
Tornabene, L. L., Moersch, J. E., McSween Jr, H. Y., McEwen, A. S., Piatek, J. L., Milam, K. A., & Christensen, P. R. (2006). Identification of large (2–10 km) rayed craters on mars in themis thermal infrared images: Implications for possible martian meteorite source regions. *Journal of Geophysical Research*, *111*(E10). https://doi.org/10.1029/2005JE002600
Tusberti, F., Pajola, M., Munaretto, G., Penasa, L., Lucchetti, A., Beccarelli, J., et al. (2024). Ice degradation and boulder size frequency distribution analysis of the fresh martian crater s1094b. *Icarus*, *423*, 116255. https://doi.org/10.1016/j.icarus.2024.116255
Wiggins, S., Johnson, B., Collins, G., Melosh, H., & Marchi, S. (2022). Widespread impact-generated porosity in early planetary crusts. *Nature Communications*, *13*(1), 4817. https://doi.org/10.1038/s41467-022-32445-3
Wiggins, S., Johnson, B. C., Bowling, T. J., Melosh, H. J., & Silber, E. A. (2019). Impact fragmentation and the development of the deep lunar megaregolith. *Journal of Geophysical Research: Planets*, *124*(4), 941–957. https://doi.org/10.1029/2018JE005757
Williams, J.-P., Bandfield, J. L., Paige, D. A., Powell, T. M., Greenhagen, B. T., Taylor, S., et al. (2018). Lunar cold spots and crater production on the moon. *Journal of Geophysical Research: Planets*, *123*(9), 2380–2392. https://doi.org/10.1029/2018JE005652
Wu, Y., Li, X., Yao, W., & Wang, S. (2017). Impact characteristics of different rocks in a pulsed laser irradiation experiment: Simulation of micrometeorite bombardment on the moon. *Journal of Geophysical Research: Planets*, *122*(10), 1956–1967. https://doi.org/10.1002/2016JE005220

21699100, 2026, 8, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JE009609 by Purdue University (West Lafayette), Wiley Online Library on [09/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License